

BUS63-8

Financial Instruments

Equities, debt, derivatives, and funds — pricing, portfolio construction, and risk management on real data

From firm-pricing to instrument-level pricing: the analytical toolkit of the Financial Markets capstone.

Albert School — 22 hours

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1 Instrument classification

This course moves from BUS43-3's firm-pricing lens (DCF, multiples, cost of capital) to instrument-level pricing, building on the Data pillar (Python, pandas) for data handling.

Definition 1 Financial instruments fall into four **families**:

- **Equities** — ownership claims on a firm's residual value; return from price appreciation and dividends, with no fixed maturity or promised payment.
- **Debt** — contractual claims to fixed or floating payments and principal at maturity; senior to equity, exposed to interest-rate and credit risk.
- **Derivatives** — contracts whose value derives from an underlying asset (futures, options, swaps); used for hedging or leveraged exposure.
- **Funds** — pooled vehicles (mutual funds, ETFs) holding a portfolio of the above, offering diversification and either active or passive management.

Definition 2 Each family has a characteristic **risk-return profile** and **investor use case**: equities for growth and equity-risk premium; debt for income and capital preservation; derivatives for hedging or speculation; funds for diversified, low-cost exposure.

Exercises

1. Classify a set of financial instruments across the four families (equities, debt, derivatives, funds) and explain the risk-return profile and typical investor use case of each.

2 Equity analysis

Definition 3 A **quantified fundamental score** rates an equity by combining three inputs computed from real market data:

- **financial ratios** — profitability, leverage, and liquidity measures from the statements (e.g. ROE, net margin, debt/equity, current ratio);
- **valuation multiples** — P/E, P/B, EV/EBITDA placing the firm relative to peers;
- **market signals** — price- and volume-based indicators (e.g. momentum, relative strength).

The score aggregates these into a single comparable figure used to rank equities.

Exercises

2. Analyse a set of equities using a quantified fundamental score built from financial ratios, valuation multiples, and market signals applied to real market data, and rank them.

3 Bond pricing mechanics

Definition 4 The **price** of a bond is the present value of its cash flows (coupons C and face value F) discounted at the **yield to maturity** (YTM) y :

$$P = \sum_{t=1}^n \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n}.$$

The YTM is the single rate that makes this present value equal to the bond's market price.

Definition 5 **Duration** measures the sensitivity of a bond's price to interest rates. **Macaulay duration** is the cash-flow-weighted average time to payment:

$$D = \left(\frac{1}{P}\right) \sum_{t=1}^n t \cdot \frac{C_t}{(1+y)^t}.$$

Modified duration $D^* = \frac{D}{1+y}$ gives the approximate percentage price change for a small yield change Δy :

$$\frac{\Delta P}{P} \approx -D^* \cdot \Delta y.$$

Definition 6 **Convexity** K is the second-order sensitivity of price to yield. It corrects the duration estimate:

$$\frac{\Delta P}{P} \approx -D^* \cdot \Delta y + \frac{1}{2}K \cdot (\Delta y)^2.$$

Because convexity is positive for standard bonds, duration alone overstates the price fall and understates the price rise.

Definition 7 **Interest-rate stress-testing** applies a range of yield shocks Δy to a real bond portfolio and revalues it using the duration-plus-convexity approximation (or full repricing) to measure the loss under each scenario.

Exercises

3. Calculate YTM, duration (Macaulay and modified), and convexity for a real bond portfolio.
4. Evaluate the portfolio's performance under interest-rate stress scenarios using the duration-and-convexity approximation.

4 Credit risk

Definition 8 **Credit risk** is the risk that a borrower fails to make promised payments. It is assessed through:

- **credit spreads** — the yield premium of a bond over a risk-free benchmark of the same maturity, compensating for default risk;
- **ratings** — agency grades (e.g. AAA to D) mapping to default probabilities;
- **default sensitivity** — how the bond's value responds to changes in default probability and recovery rate.

Definition 9 A bond's yield decomposes into a **risk-free rate** plus a **credit spread**. Fixed-income value therefore moves with both **interest-rate risk** (the benchmark rate) and **credit risk** (the spread): a position can lose value from a rate rise, a spread widening, or both, and the two can move together in stressed markets.

Exercises

5. Assess the credit risk of a bond via spreads, ratings, and default sensitivity.
6. Explain how credit risk interacts with interest-rate risk in fixed-income valuation.

5 Derivatives

Definition 10 A **futures contract** obliges its holder to buy or sell an underlying asset at a fixed price on a future date. Under **arbitrage (cost-of-carry) pricing**, the fair futures price is the spot price compounded at the risk-free rate net of any income or carrying cost:

$$F_0 = S_0 \cdot e^{rT}$$

(continuous compounding, no income). Any deviation permits a riskless arbitrage.

Definition 11 An **option** gives the right, not the obligation, to buy (**call**) or sell (**put**) the underlying at a **strike price** K before or at expiry. The **Black-Scholes** model prices a European option from the spot S , strike K , time to expiry T , risk-free rate r , and volatility σ ; intuitively, option value rises with volatility and time, since both increase the chance of finishing in-the-money while losses remain capped at the premium.

Definition 12 The **Greeks** measure an option's sensitivities:

- **delta** — change in option value per unit change in the underlying price;
- **gamma** — change in delta per unit change in the underlying price;
- **vega** — change in option value per unit change in volatility.

Definition 13 A **swap** exchanges two streams of payments. An **interest-rate swap** exchanges fixed for floating interest on a notional principal; a **currency swap** exchanges principal and interest in one currency for those in another. Swaps are used to transform or hedge an exposure's rate or currency profile.

Exercises

7. Explain the economic logic of futures (arbitrage pricing), options (calls, puts, Black-Scholes intuition, the Greeks delta, gamma, vega), and interest-rate and currency swaps.

6 Hedging strategy design

Definition 14 A **hedge** is a derivative position taken to offset the risk of an existing exposure. Designing one on real exposure data requires three steps:

- **identify the exposure** — the asset, rate, or currency whose movement creates risk, and its direction and size;
- **select the instrument** — the future, option, or swap whose payoff offsets that movement;
- **calibrate the position** — size the hedge (e.g. via the hedge ratio, often the exposure's delta or duration) so the offsetting payoff matches the exposure.

Exercises

8. Design a hedging strategy for a real exposure: identify the exposure, select the appropriate instrument, and calibrate (size) the position on the exposure data.

7 Funds and ETFs

Definition 15 A **fund** (mutual fund or ETF) pools capital into a managed portfolio. Its performance is judged **relative to a benchmark** index:

- **active return** — the fund return minus the benchmark return;
- **tracking error** — the standard deviation of the active return, measuring how closely the fund follows its benchmark.

Definition 16 **Active vs passive efficiency** compares funds that seek to beat a benchmark (active, higher fees, non-zero tracking error) against those that replicate it (passive, low fees, low tracking error). **Fund-selection analytics** evaluate a fund on benchmark-relative return, tracking error, and cost on historical data.

Exercises

9. Evaluate fund and ETF performance against benchmarks: compute tracking error and active return on historical data.

10. Interpret active-vs-passive efficiency and apply fund-selection analytics to choose among funds.

8 Multi-asset portfolio evaluation

Definition 17 A **multi-asset portfolio** is evaluated on historical data with:

- **volatility** — the standard deviation of portfolio returns;

- **Sharpe ratio** — excess return per unit of risk,

$$\text{Sharpe} = \frac{R_p - R_f}{\sigma_p};$$

- **maximum drawdown** — the largest peak-to-trough decline in portfolio value over the period;
- **cross-asset correlations** — the pairwise correlation of asset returns, which drives the diversification benefit.

Definition 18 The **diversification signal** is read from the correlation structure: combining assets with low or negative correlation reduces portfolio volatility below the weighted average of the individual volatilities.

Exercises

11. Evaluate a multi-asset portfolio on historical data: compute volatility, Sharpe ratio, maximum drawdown, and cross-asset correlations.
12. Interpret the diversification signal from the correlation structure.

9 Portfolio construction

Definition 19 Portfolio construction assembles a portfolio aligned with a defined **risk profile** through: **selection** (which assets), **sizing** (the weights), **diversification** (spreading risk across low-correlation assets), and **risk-profile alignment** (matching the resulting volatility to the investor's tolerance).

Definition 20 Mean-variance optimisation chooses portfolio weights to minimise variance for a target expected return (or maximise return for a target variance), tracing the **efficient frontier** of portfolios with the best return for each level of risk. The inputs are the assets' expected returns, volatilities, and correlations.

Common pitfall Mean-variance optimisation has **practical limits** that must be stated when it is applied: it is highly sensitive to the estimated expected returns and correlations, which are measured with error from historical data; small input changes produce large, unstable weight shifts, often concentrating the portfolio in a few assets. The output is a guide, not a prescription.

Exercises

13. Construct and justify a multi-asset portfolio aligned with a defined risk profile, applying mean-variance optimisation logic.
14. Acknowledge explicitly the practical limits of mean-variance optimisation in the construction.

10 Communicating an investment recommendation

Definition 21 An **investment recommendation** presents a constructed portfolio to a **non-technical audience** using:

- **data visualisation** in Python / matplotlib — charts of allocation, historical performance, and risk metrics;
- **narrative structure** — a written rationale covering return expectations, risk management, and the data sources used.

The recommendation states the portfolio selected, how it is sized, and why, translating the quantitative analysis into a decision the audience can act on.

Exercises

15. Communicate a data-driven investment recommendation to a non-technical audience using Python / matplotlib visualisation and narrative structure — selecting, sizing, and presenting a portfolio with a written rationale covering return expectations, risk management, and data sources used.

This course equips the Financial Markets capstone with an instrument-level pricing toolkit — equity scoring, bond duration and convexity, credit spreads, the Greeks, and mean-variance portfolio construction — applied to real data and paired with BUS63-9 Investment Philosophies, which asks when to deploy each approach.